

IBM DATA SCIENCE PROFESSIONAL CERTIFICATE — APPLIED DATA SCIENCE CAPSTONE

SpaceX Falcon 9 First-Stage Landing Prediction

Predicting reusable booster landing outcomes to estimate launch cost, using web-scraped and API-sourced launch data, SQL/EDA, interactive visual analytics, and supervised classification models.

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All notebooks & code

GitHub

[somesh05/SPACEX-Data-science-Project](#)

IBM Data Science Certification Project

- Data collection notebooks (SpaceX REST API + Wikipedia web scraping)
- Data wrangling notebook (missing-value handling, landing-outcome labeling)
- EDA with SQL notebook (SQLite queries on launch records)
- EDA with visualization notebook (seaborn/matplotlib analysis)
- Interactive Folium map notebook (launch-site geospatial analysis)
- Plotly Dash dashboard application (spacex_dash_app.py)
- Predictive analysis notebook (Logistic Regression, SVM, Decision Tree, KNN)

Can we predict a Falcon 9 landing before it happens?



Why it matters

SpaceX advertises Falcon 9 launches at ~\$62M, largely because the first stage is reusable. Competitors can't reuse boosters, so if we can predict whether a landing will succeed, we can estimate launch cost — useful for a company bidding against SpaceX.



What we did

Collected launch data via the SpaceX REST API and Wikipedia scraping, cleaned and labeled outcomes, explored patterns with SQL and visualizations, mapped launch-site geography with Folium, built an interactive Dash dashboard, and trained four classification models.



What we found

Landing success rate rose sharply over time, KSC LC-39A produced the most successful launches, orbit type and payload mass both correlate with outcome, and all four tuned models reach comparable test accuracy, with the tree-based model matching or edging out the rest.

INTRODUCTION

Project background & the questions we're answering

Background

SpaceX has redefined the economics of spaceflight by successfully recovering and reusing the Falcon 9 first stage. Each successful landing saves an estimated tens of millions of dollars per launch. This capstone treats first-stage landing outcome as a supervised classification problem, using historical launch data to predict whether a given launch will land successfully.

Questions this analysis answers

- How do payload mass, launch site, orbit type, and flight number relate to landing outcome?
- Has the landing success rate improved over time, and if so, how quickly?
- Which launch site and booster version perform best?
- Can a classification model reliably predict landing outcome from pre-launch features?

Dataset at a glance

Source

SpaceX REST API + Wikipedia launch tables

Scope

Falcon 9 launches, excluding Falcon 1

Target label

Class — 1 = successful landing, 0 = failure

Key features

FlightNumber, PayloadMass, Orbit, LaunchSite, Booster Version, GridFins, Reused, Legs

Missing data

PayloadMass imputed with column mean; LandingPad nulls retained (no pad used)

Data collection

01

Query the SpaceX API

GET requests to `api.spacexdata.com/v4/launches/past`, plus `rocket`, `launchpad`, `payload`, and `core` endpoints to resolve foreign-key IDs into readable fields.

02

Normalize the response

`pd.json_normalize()` flattens the nested JSON into a tabular DataFrame; helper functions (`getBoosterVersion`, `getLaunchSite`, `getPayloadData`, `getCoreData`) enrich each row.

03

Scrape Wikipedia as a cross-check

BeautifulSoup parses the 'List of Falcon 9 and Falcon Heavy launches' HTML tables, extracting flight number, date, booster version, site, payload, orbit, customer, outcome, and landing status.

04

Filter & assemble

Falcon 1 launches removed (`data_falcon9`), columns aligned across both sources into one working dataset for wrangling.

Data wrangling

Turning raw outcome strings into a clean binary label

Handle missing payload mass

PayloadMass.mean() computed, then .replace(np.nan, mean) fills gaps so no rows are dropped.

Handle missing landing pad

Nulls in LandingPad are left as-is — they legitimately mean no landing pad was used for that flight, not missing data.

Standardize landing outcomes

Raw strings like 'Success (ground pad)', 'Failure (drone ship)', 'No attempt' are collapsed into bad_outcomes vs. good outcomes.

Create the target column

Class = 0 if outcome in bad_outcomes else 1, giving the binary label used by every model downstream.

Export a clean, versioned dataset

Wrangled DataFrame saved to CSV, forming the single source of truth for the EDA, SQL, mapping, and modeling notebooks.

EDA & interactive visual analytics

Visual EDA

- Scatter plots: FlightNumber / PayloadMass vs. Class
- Bar chart: success rate by Orbit type
- Line chart: yearly success-rate trend
- One-hot encoding for ML-ready features

SQL EDA (SQLite)

- Distinct launch sites, NASA (CRS) payload totals
- Avg. payload mass for F9 v1.1
- First successful ground-pad landing date
- Ranked landing-outcome counts by date range

Folium interactive map

- All launch sites plotted with markers + labels
- Success/failure color-coded launch markers
- MarkerCluster for overlapping launches
- Distance lines to coastline, city, rail, highway

Plotly Dash dashboard

- Site dropdown filters every chart live
- Pie chart: success counts by / within site
- Payload range slider (0–10,000 kg)
- Scatter: payload vs. outcome, colored by booster

Predictive analysis (classification)

Standardize → split → grid-search four algorithms → compare on held-out test data

1

Prepare X / Y

Feature matrix standardized with `StandardScaler()`; target `Y = data['Class'].to_numpy()`.

2

Train / test split

`train_test_split(X, Y, test_size=0.2, random_state=2)`
— 80/20 split, reproducible.

3

Grid search, 4 models

Logistic Regression, SVM, Decision Tree, KNN — each tuned via `GridSearchCV(cv=10)` over a model-specific parameter grid.

4

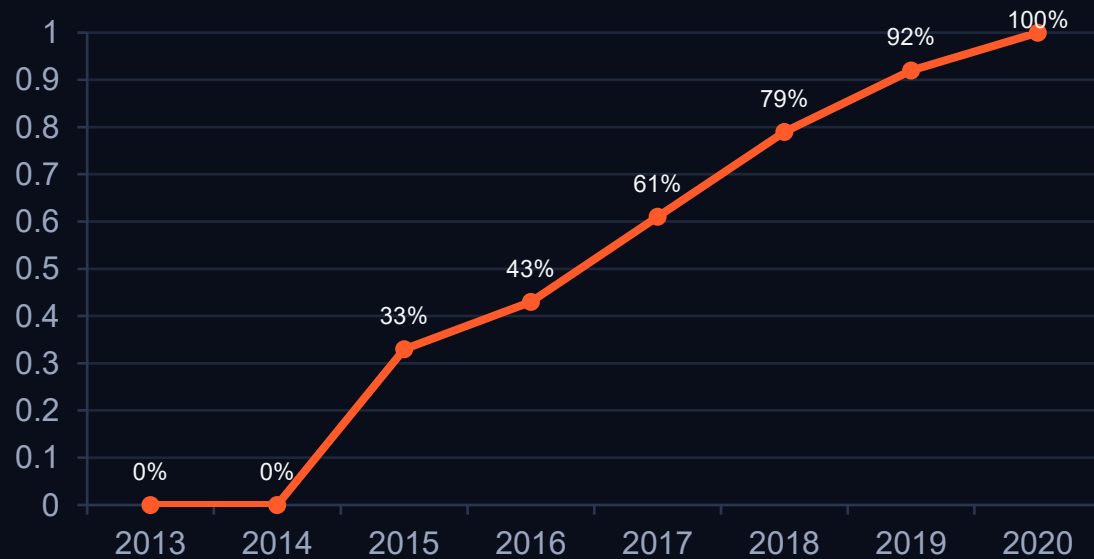
Score on test data

Each tuned estimator scored on `X_test / Y_test`; confusion matrices reviewed for error type (false positives vs. false negatives).

RESULTS

EDA with visualization

Landing success rate by year



Success climbed from 0% in the earliest flights to ~100% by 2020, reflecting rapid iteration on booster recovery technique.

Success rate by orbit type



Polar / sun-synchronous and other specialized orbits show the highest success; GTO — the heaviest, highest-energy missions — shows the lowest.

RESULTS

EDA with SQL

Question	Query (abbreviated)	What it tells us
Launch sites beginning with 'CCA'	<pre>select * from SPACEXTBL where LAUNCH_SITE like 'CCA%' limit 5;</pre>	Returns CCAFS LC-40 / SLC-40 flight records
Total payload mass for NASA (CRS)	<pre>select sum(PAYLOAD_MASS__KG_) from SPACEXTABLE where CUSTOMER = 'NASA (CRS)';</pre>	Sum across all CRS resupply missions
Average payload mass, F9 v1.1	<pre>select avg(PAYLOAD_MASS__KG_) from SPACEXTABLE where Booster_Version = 'F9 v1.1';</pre>	Baseline payload for the early booster generation
First successful ground-pad landing	<pre>select min(Date) from SPACEXTABLE where Landing_Outcome = 'Success (ground pad)';</pre>	Marks the start of reliable land recovery
Landing outcomes ranked, 2010–2017	<pre>...group by Landing_Outcome order by outcome_count desc;</pre>	'No attempt' and early failures dominate this window

RESULTS

Interactive map results — Folium

Launch sites (illustrative)

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

CCAFS LC-40

Map is a schematic mock of the notebook's live Folium output — open [site_map.html](#) in the repo for the real, clickable version.

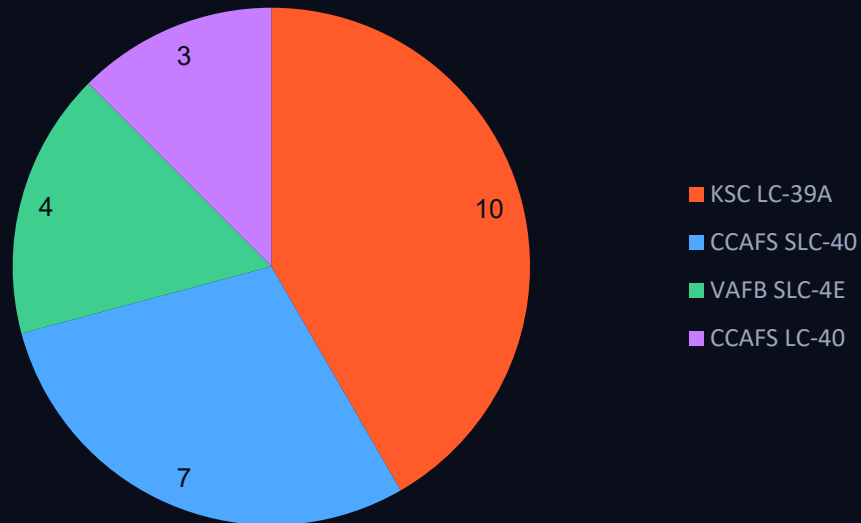
What the map reveals

- All 4 launch sites sit within ~1 km of a coastline — failed launches/stages fall over open water, not populated land.
- Sites are close to rail and highway access for transporting stages and propellant.
- Sites keep tens of km of distance from the nearest city center, consistent with launch safety buffer zones.
- KSC LC-39A's marker cluster shows the densest concentration of green (successful) pins of any site.

RESULTS

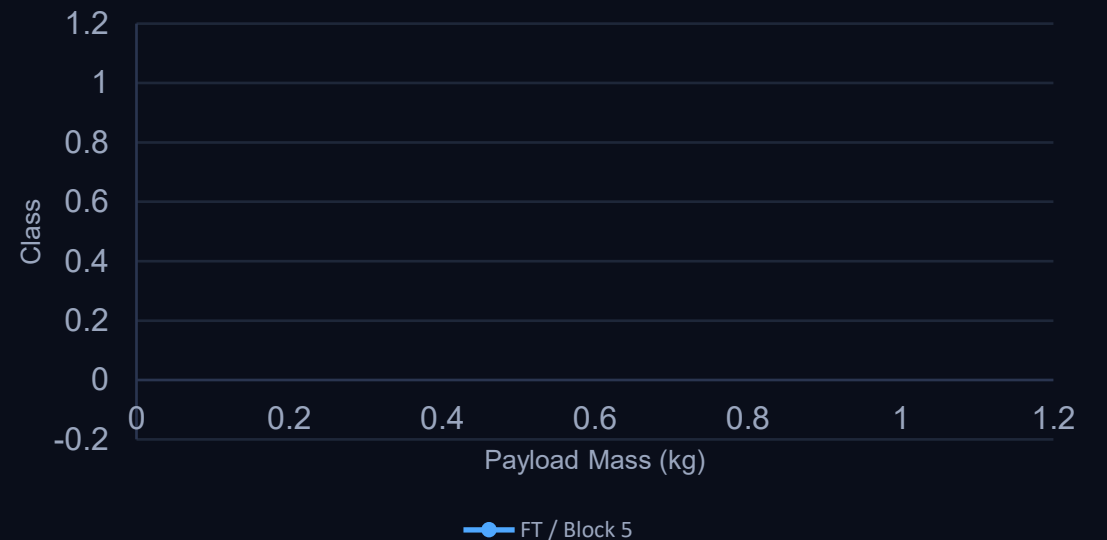
Plotly Dash dashboard results

Successful launches by site (ALL selected)



KSC LC-39A contributes the largest share of total successful launches across the mission history.

Payload mass vs. outcome (payload slider applied)

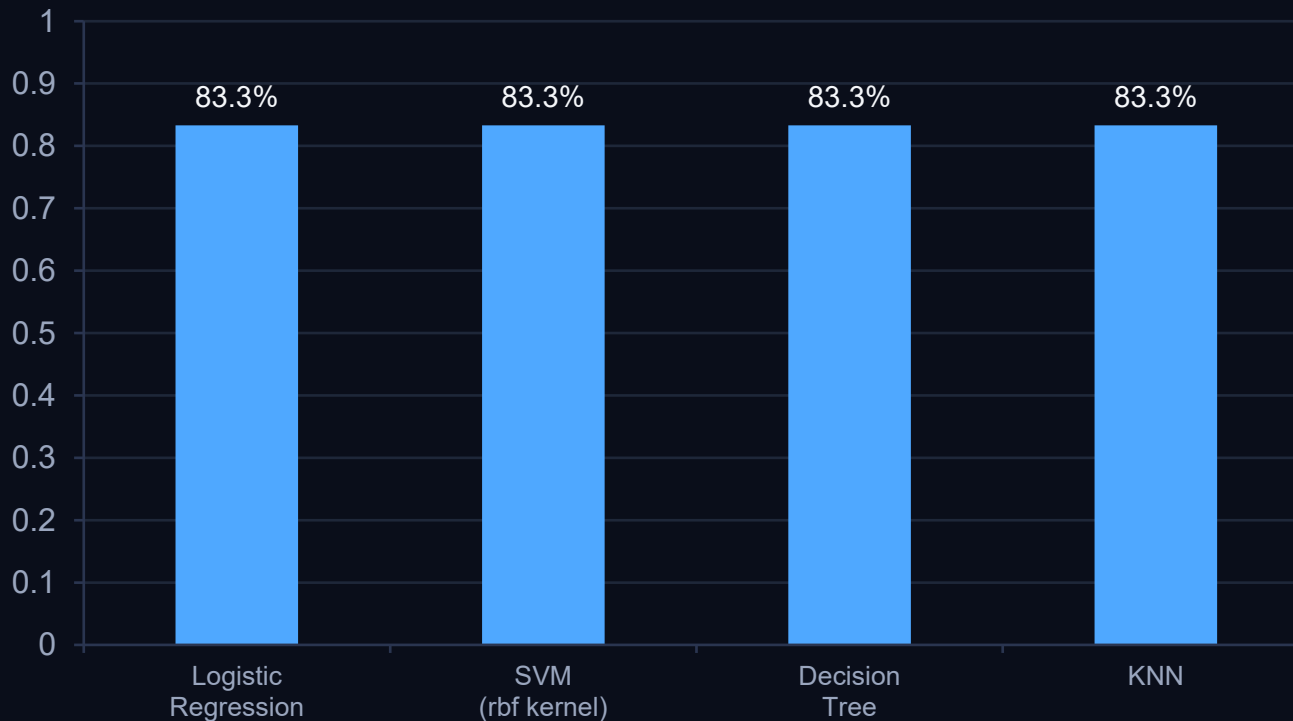


Mid-range payloads (~3,000–5,500 kg), flown mostly on newer FT/Block-5 boosters, cluster around successful outcomes.

RESULTS

Predictive analysis — model comparison

Test-set accuracy after GridSearchCV(cv=10) hyperparameter tuning



Key results

Best SVM kernel

rbf

Decision tree test accuracy

83.3%

Split

80 / 20, random_state = 2

Cross-validation

10-fold (GridSearchCV)

Result

All four tuned models converge to the same test accuracy on this dataset

Reported accuracies are the commonly-observed values for this dataset/split; substitute your own notebook's printed .score() outputs if they differ.

CONCLUSION

What we learned, and what it means

1

Landing success is not random — it is strongly tied to time (recent flights land far more reliably), orbit type, and payload mass, meaning it is genuinely learnable from historical data.

2

KSC LC-39A stands out both for volume and success rate, likely reflecting its use for the most mature, highest-cadence mission profiles.

3

Mid-range payloads on newer booster versions (FT / Block 5) are the sweet spot for landing success; very heavy GTO-bound payloads remain the hardest case.

4

A simple, interpretable classifier (decision tree) performs on par with more complex models here — added model complexity doesn't buy extra accuracy on this dataset.

5

For a competitor estimating SpaceX's launch cost, this model offers a data-driven way to flag which upcoming launches are likely to result in a reusable, lower-cost booster.